PHYSICS

Stage 6

Module 1: Kinematics

Motion in a Straight Line

Teacher: Joshua Griffiths

Exam Equivalent Time: 57 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. PHYSICS, M1 2013 HSC 4 MC

Students performed an investigation to determine the initial velocity of a projectile.

Which row correctly identifies a hazard of this investigation and a related precaution?

	Hazard	*Safety precaution*
A.	flying projectile	wearing safety glasses
B.	range of projectile	measuring the range with a tape measure
C.	enclosed shoes	limiting the range of the projectile
D.	safety glasses	flying projectile

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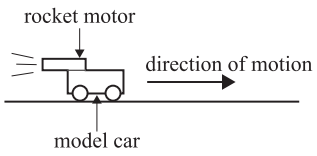
2. PHYSICS, M1 2017 VCE 9*

A model car of mass 2.0 kg is propelled from rest by a rocket motor that applies a constant horizontal force of 4.0 N, as shown below. Assume that friction is negligible.



With the same rocket motor, the car accelerates from rest for 10 seconds.

Calculate the the final speed of the model car? ***(2 marks)***

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3. PHYSICS, M1 EQ-Bank 8

A truck travelling in a straight line with a speed of 60 ms^{-1} slows down and comes to rest over a period of 20 seconds.

a. Calculate the change in velocity of the truck. ***(2 marks)***

b. Determine the average acceleration of the truck across the 20 seconds. ***(2 marks)***

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4. PHYSICS, M1 EQ-Bank 1

A ball is thrown vertically upwards from ground level, it gains 50 metres vertically and then falls back to the ground.

i. Determine the initial velocity, in metres per second, that the ball was thrown. Give your answer correct to one decimal place. ***(2 marks)***

ii. Calculate the total time of flight of the ball, in seconds, giving your answer correct to two decimal places. ***(2 marks)***

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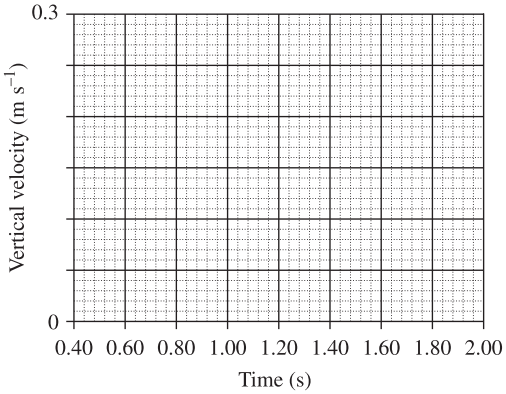
5. PHYSICS, M1 2013 HSC 22

This set of data was obtained from a motion investigation to determine the acceleration due to gravity on a planet other than Earth.

Time (s)	Vertical velocity (m s^{-1})
0.60	0.02
1.00	0.09
1.20	0.12
1.40	0.17
1.80	0.23

Plot the data from the table, and then calculate the acceleration. ***(3 marks)***



6. PHYSICS, M1 EQ-Bank 10

A motorbike is travelling to the east at 50 ms^{-1} when it passes a car travelling west at 60 ms^{-1}.

Calculate the velocity of the motorbike relative to the car. ***(2 marks)***

7. PHYSICS, M1 EQ-Bank 11

A plane is travelling at 315 ms^{-1} north when it passes through a dense cloud and slows down to a velocity of 265 ms^{-1} for safety precautions.

The plane did not change direction and travelled 2.5 km while it was slowing down.

Using north as the positive direction for all calculations, determine:

a. the change in velocity of the plane. ***(1 mark)***

b. the plane's acceleration. ***(2 marks)***

c. the time over which the plane slowed down. ***(2 marks)***

8. PHYSICS, M1 EQ-Bank 2

A cricket ball is hit vertically upwards from ground level, it gains 70 metres vertically and then falls back to the ground.

At what time(s) will the ball be 40 metres above the ground? Ignore any air resistance, giving your answer in seconds, correct to two decimal places. ***(4 marks)***

9. PHYSICS, M1 EQ-Bank 7

There has been an error, and the content for this question is not available. Please notify support.

10. PHYSICS, M1 EQ-Bank 9

Plane A is flying due north at 300 kmh^{-1} when it measures the velocity of plane B flying due south to be 750 kmh^{-1}.

Calculate the velocity of plane B as measured by the pilots on plane B? ***(3 marks)***

11. PHYSICS, M1 2012 HSC 21

a. Outline a first-hand investigation that could be performed to measure a value for acceleration due to gravity. ***(3 marks)***

b. How would you assess the accuracy of the result of the investigation? ***(1 mark)***

c. How would you increase the reliability of the data collected? ***(1 mark)***

d. How would you assess the reliability of the data collected? ***(1 mark)***

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Worked Solutions

1. PHYSICS, M1 2013 HSC 4 MC

→ A hazard is a danger and the only danger in the answer options is the flying projectile.

→ Wearing safety glasses will protect students eyes from being hit from the projectile.

$\Rightarrow A$

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2. PHYSICS, M1 2017 VCE 9*

$$a = \frac{F}{m} = \frac{4.0}{2.0} = 2\ \text{ms}^{-2}$$

$$\begin{aligned} v &= u + at \\ &= 0 + 2 \times 10 \\ &= 20\ \text{ms}^{-1} \end{aligned}$$

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3. PHYSICS, M1 EQ-Bank 8

a. $$\begin{aligned} \Delta v &= v - u \\ &= 0 - 60 \\ &= -60\ \text{ms}^{-1} \end{aligned}$$

b. $$\begin{aligned} a &= \frac{v-u}{t} \\ &= \frac{-60}{20} \\ &= -3\ \text{ms}^{-2} \end{aligned}$$

Worked Solutions

4. PHYSICS, M1 EQ-Bank 1

i. At 50 m $\Rightarrow v = 0\ \text{ms}^{-1}$

$$\begin{aligned} v^2 &= u^2 + 2as \\ 0^2 &= u^2 + 2 \times -9.8 \times 50 \\ u^2 &= 980 \\ u &= 31.3\ \text{ms}^{-1} \end{aligned}$$

ii. Time to highest point $= \frac{1}{2}$ time of flight

$$\begin{aligned} v &= u + at \\ 0 &= 31.3 - 9.8t \\ 9.8t &= 31.3 \\ t &= 3.194\ \text{s} \end{aligned}$$

Total time of flight $= 2 \times 3.194 = 6.39$ s (2 d.p.)

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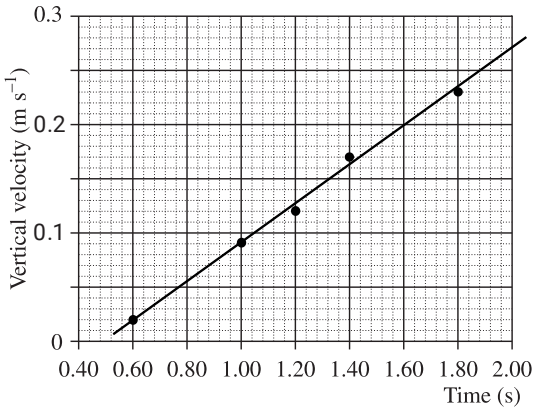
5. PHYSICS, M1 2013 HSC 22



$$\begin{aligned} \text{Acceleration} &= \frac{0.2 - 0.04}{1.6 - 0.72} \\ &= \frac{0.16}{0.88} \\ &= 0.182\ \text{ms}^{-2} \end{aligned}$$

6. PHYSICS, M1 EQ-Bank 10

Let east be the positive direction and west be the negative direction.

$$\begin{aligned} v_{\text{m rel c}} &= v_{\text{m}} - v_{\text{c}} \\ &= 50 - (-60) \\ &= 110\ \text{ms}^{-1} \end{aligned}$$

Therefore, the velocity of the motorbike relative to the car is 110 ms^{-1} east.

7. PHYSICS, M1 EQ-Bank 11

a. $$\begin{aligned} \Delta v &= v - u \\ &= 265 - 315 \\ &= -50\ \text{ms}^{-1} \\ &= 50\ \text{ms}^{-1}\ \text{south} \end{aligned}$$

b. Using $v^2 = u^2 + 2as$ (time is not given):

$$\begin{aligned} a &= \frac{v^2 - u^2}{2s} \\ &= \frac{(265)^2 - (315)^2}{2 \times 2500} \\ &= -5.8\ \text{ms}^{-2} \\ &= 5.8\ \text{ms}^{-2} \end{aligned}$$ to the south.

c. Using $v = u + at$:

$$\begin{aligned} t &= \frac{v - u}{a} \\ &= \frac{265 - 315}{-5.8} \\ &= 8.62\ \text{s} \end{aligned}$$

8. PHYSICS, M1 EQ-Bank 2

$$\begin{aligned} &\text{At } 70\ \text{m} \Rightarrow v = 0\ \text{ms}^{-1}: \\ &v^2 = u^2 + 2as \\ &0^2 = u^2 + 2 \times -9.8 \times 70 \\ &u^2 = 1372 \\ &u = 37.04\ \text{ms}^{-1} \end{aligned}$$

$$\begin{aligned} &\text{Find } t \text{ when } s = 40: \\ &s = ut + \frac{1}{2}at^2 \\ &40 = 37.04t + \frac{1}{2} \times -9.8 \times t^2 \\ &4.9t^2 - 37.04t + 40 = 0 \\ &t = \frac{37.04 \pm \sqrt{(-37.04)^2 - 4 \times 4.9 \times 40}}{2 \times 4.9} \\ &= 6.25\ \text{s and } 1.31\ \text{s} \end{aligned}$$

9. PHYSICS, M1 EQ-Bank 7

$s = 50,\ a = 9.8,\ u = -15$

Downwards $\Rightarrow$ positive direction

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ 50 &= -15t + \frac{1}{2} \times 9.8 \times t^2 \\ 0 &= 4.9t^2 - 15t - 50 \\ t &= \frac{15 \pm \sqrt{225 - 4 \times 9.8 \times -50}}{2 \times 4.9} \\ &= 5.07, -2.01 \end{aligned}$$

Time for the pen to fall = 5.07 seconds $(t > 0)$.

10. PHYSICS, M1 EQ-Bank 9

Let north be the positive direction and south be the negative direction.

$$\begin{aligned} v_{\text{B rel A}} &= v_{\text{B}} - v_{\text{A}} \\ -750 &= v_B - 300 \\ v_B &= -450 \\ &= 450\ \text{kmh}^{-1}\ \text{south} \end{aligned}$$

11. PHYSICS, M1 2012 HSC 21

a. Timing of a falling mass.

♦ Mean mark (a) 55%.

→ Set up an electronic and automatic timing system with sensors to detect the presence of a small falling metal ball.

→ Heights for the ball should be set up between 0.2 m to 1 m with intervals every 0.2 m. To increase the reliability of the results, multiple trials should be conducted at each height and the average falling time for each height should be calculated which can then be used to graph the data.

→ The results should be plotted on a graph of height vs time^2. This uses the equation $s = ut + \frac{1}{2}at^2$ where $u = 0$ which becomes $s = \frac{1}{2}at^2$.

→ After plotting the data, the acceleration due to gravity, a, can be calculated using $a = \frac{2s}{t^2}$, which will make it equal to 2 × the gradient of the line of best fit.

b. → Look up known value on a reliable website (e.g. National Measurement Institute).

♦♦♦ Mean mark (b) 26%.

→ Ensure the value is for the location of the experiment (it can differ slightly).

→ Compare the known value to the value determined experimentally and the closer they are, the greater the accuracy of the experiment.

c. → Conduct multiple trials at each height.

→ Use the average of the calculations as stated in the method above.

♦♦♦ Mean mark (d) 7%.

d. → Compare the values obtained at a single height.

→ If there is a large variation in the calculations conducted at the same height, the data collected is less reliable.

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